

Soft Spaces Phase 4

v40.6b IBM Heron Hardware Confirmation - Summary

IBM Quantum | ibm_kingston | Heron revision 2

Primary result: REAL outperformed the matched NULL control on physical IBM Heron hardware. All four predeclared Precision@8 decision criteria passed.

1. Purpose of the experiment

v40.6b was the first Soft Spaces Phase-4 REAL-vs-NULL experiment executed on a physical IBM Quantum processor. The goal was deliberately narrower than a full native implementation of the complete Soft Spaces Hamiltonian/projector model. Instead, the frozen Phase-3 source information was encoded into explicit quantum circuits, passed through real Heron hardware, measured with finite shots, and then used for the same ranked-search endpoint previously tested in Qiskit/Aer.

The scientific question was therefore: does the REAL-specific ranking advantage survive the transition from a simulated noisy measurement channel to actual quantum hardware?

2. Frozen experimental design

Backend	ibm_kingston
Processor	IBM Heron, revision 2
Logical qubits	8
Frozen physical layout	[47, 48, 46, 49, 45, 50, 44, 38]
Seed ensemble	25043000 ... 25043011 (12 independent seeds)
Models	REAL and matched NULL
Primary endpoint	Precision@8
Secondary endpoints	Target evaluations and NDCG@8
Two-qubit checkpoint	20 logical CX identity-loading operations
Transpiler optimization	Level 0
Mitigation	None in the primary comparison

Budget adaptation. The original frozen protocol specified 4096 shots per circuit. To remain within the available Open-plan QPU budget, v40.6b used 2304 shots per circuit. The seed ensemble, REAL/NULL definition, K=8 endpoint, CX=20 checkpoint, backend, frozen physical layout, raw/no-mitigation comparison, and ranking/statistical endpoints were otherwise unchanged. Accordingly, v40.6b should be described as a budget-adapted hardware confirmation rather than the untouched 4096-shot primary confirmation.

3. Hardware execution

Candidate circuits	592
Shots per circuit	2304
Total executions	1,363,968
Transpiled circuit depth	88 for every circuit
Native 2Q gates	20 for every circuit
Submitted	Yes

IBM Runtime job ID	daknuamlg97s73c7ur40
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The identical transpiled depth and identical native two-qubit gate count across all candidate circuits are important. They indicate that REAL and NULL were compared under a uniform hardware-routing burden rather than under systematically different circuit structures.

4. Primary hardware result

Metric	REAL	NULL
Mean Precision@8	0.96875	0.82292
Mean target evaluations	8.25	10.83

Mean delta Precision@8 = +0.14583 (approximately +14.6 percentage points in favor of REAL).
95% bootstrap confidence interval: [+0.08333, +0.20833]

One-sided exact sign-test: $p = 0.001953$

One-sided Wilcoxon signed-rank test: $p = 0.003899$

5. Predeclared decision gates

Criterion	Requirement	Result
C1	Mean delta Precision@8 > 0	PASS
C2	95% bootstrap CI lower bound > 0	PASS
C3	Exact sign-test $p < 0.05$	PASS
C4	Wilcoxon $p < 0.05$	PASS

FINAL v40.6b HARDWARE DECISION: PASS

6. Seed-level pattern

Across the 12 independent seeds, REAL had a positive Precision@8 advantage in 9 seeds, tied NULL in 3 seeds, and was never lower than NULL. One strong example was seed 25043006, where REAL achieved Precision@8 = 1.000 versus NULL = 0.625, giving delta = +0.375.

7. Secondary ranking metrics

Mean evaluations saved by REAL	2.583
95% bootstrap CI, evaluations saved	[1.417, 3.750]
Mean delta NDCG@8	+0.0985
95% bootstrap CI, delta NDCG@8	[+0.0478, +0.1562]

These secondary measures point in the same direction as Precision@8: the REAL ranking tended to locate relevant targets with fewer evaluations and produced better ranking quality than the matched NULL control.

8. Comparison with the previous Aer result

Environment	Delta Precision@8	Interpretation
Qiskit/Aer v40.3b	+0.15328	Explicit noisy simulated measurement channel
IBM Heron v40.6b	+0.14583	Physical quantum hardware

The physical-hardware effect size remained close to the earlier Aer result rather than collapsing under real device noise. That agreement is an important consistency check, although it should not be interpreted as evidence of a full native quantum implementation.

9. Actual IBM Heron execution time

Job created	2026-09-15 17:23:54 UTC
QPU running	2026-09-15 20:46:35 UTC
Job finished	2026-09-15 20:53:21 UTC
IBM qpu_charge_time_seconds	362 s = 6 min 2 s
IBM circuits_execution_time_ns	359,027,534,080 ns = 359.03 s = 5 min 59.0 s

The long wall-clock delay was therefore almost entirely queue/wait time. The actual hardware execution was approximately six minutes. The preflight estimate had predicted about 351.8 seconds (5.86 minutes), which was only about 7 seconds below the actual billed QPU time.

10. Scientific interpretation

The strongest defensible conclusion from v40.6b is that the REAL-specific ranked-search advantage survives an explicit physical measurement channel on IBM Heron hardware. The advantage remains positive across the seed ensemble, passes the predeclared statistical gates, and is supported by secondary ranking metrics.

What this result does establish:

- The v40.3b-style source-to-ranking measurement channel can be executed on real IBM Heron hardware.
- The REAL-vs-NULL ranking advantage survives real gate noise, readout noise, and finite-shot sampling.
- The observed hardware effect size is close to the earlier Qiskit/Aer result.

What this result does not establish:

- A full native quantum implementation of the complete Soft Spaces Hamiltonian/projector calculation.
- Quantum computational advantage.
- The untouched 4096-shot frozen primary confirmation, because v40.6b used the explicit 2304-shot budget adaptation.

11. Short-form conclusion

Soft Spaces v40.6b was executed on a physical IBM Heron processor using 592 circuits and 1.36 million circuit executions. REAL achieved mean Precision@8 = 0.969 versus 0.823 for the matched NULL control, giving delta Precision@8 = +0.146. The 95% bootstrap confidence interval remained entirely above zero, both paired statistical tests were significant, and all four predeclared decision gates passed. The actual IBM QPU charge time was 362 seconds, or 6 minutes 2 seconds. The result therefore provides direct hardware evidence that the REAL-specific ranked-search advantage survives the transition from a simulated noisy circuit channel to physical IBM Heron hardware, while remaining distinct from a full Hamiltonian-native quantum implementation.